

Xin Huang

Hong Kong

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Research Interests

Continual Learning; Stability–Plasticity Theory; Non-stationary Learning and Optimization; Reliable ML Experimentation

Education

The University of Hong Kong

M.Sc. in Computer Science, School of Computing and Data Science
GPA to date: 3.51/4.30; Pass with Distinction (Top 10% in Msc(Comp))

Hong Kong

Sep 2025 – Expected Nov 2026

The University of Hong Kong

B.Sc. in Mathematics, Minor in Computer Science
GPA: 3.64/4.30; First Class Honours; Dean's Honours List (2023, 2024)

Hong Kong

Sep 2021 – Jul 2025

Selected Manuscript

Xin Huang, Yujing Liu, Fei Zhu, Zhixin Liu, Dacheng Tao

“Stability–Plasticity Theory for Continual Learning under Drifting Task Optima.”

First-author manuscript under review at *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. Manuscript available upon request.

Research and Technical Experience

Research Intern – AI Models & Systems

Hong Kong

Supervisor: Fei Zhu, Hong Kong Institute of Science & Innovation, Chinese Academy of Sciences (CAIR-CAS)
Sep 2025 – May 2026

- Formulated continual learning under drifting task optima as a coupled tracking problem with a history-dependent importance metric as part of the learner state.
- Derived tracking guarantees and a loss decomposition separating current-task tracking error, intrinsic task discrepancy, and a learned-geometry interaction term.
- Analyzed task-order effects using analytical examples and experiments on synthetic data and CIFAR benchmarks.

Algorithm Engineer Intern – LLM Agent Optimization

Hangzhou, China

Ant Group, Credit Business Unit

Jun 2026 – Present

- Contribute to the design and implementation of Auto-Research, a framework for iterative ML experimentation, focusing on experiment candidate generation, orchestration, evidence tracking, and evaluation.
- Apply the framework to recommendation-model experiments for credit-related tasks and conduct controlled offline evaluations of candidate model changes.
- Analyze long-horizon agent traces and failure cases to inform system revisions and evaluation procedures, with findings documented in technical research materials.

Research Assistant

Hong Kong

Supervisor: Feng Tian, HKU Business School, The University of Hong Kong

Nov 2024 – Mar 2025

- Derived and checked proofs of theorems and lemmas in operations management models involving constrained contracts and strategic decisions.
- Analyzed resource-allocation and supply-chain formulations using linear programming, KKT-based analysis, and gradient methods.

Algorithm Research Intern

Alberta, Canada

Supervisor: Bin Han, Department of Mathematical and Statistical Sciences, University of Alberta
Jun 2024 – Aug 2024

- Implemented and evaluated Kolmogorov–Arnold Network variants for recommendation, comparing B-spline, Fourier, and wavelet basis functions.
- Explored hypergraph-enhanced architectures and benchmarked them against graph-based recommendation baselines, including NGCF and LightGCN.

Honours and Awards

- HKU Entrance Full Scholarship (2021–2022).
- HKU Full Scholarship for Outstanding Academic Talents (2022–2025); Dean's Honours List (2023, 2024).